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NIKI — Cognitive Research & Invention

Research mandates, algorithm proposals, testing, repair and qualified knowledge retention

BTG-LIB-NIKI-SG · Edition 1.0 · Audience: Research sponsors, institutional reviewers, technical project owners and oversight teams · Issued 23 October 2026

Publication boundary: This is an explanation of the system and its public operating model, not a source release, engineering specification, certification or promise of live service availability. Product status is stated with the limitations below.

Executive overview

NIKI is BTG's autonomous research and invention programme. Its ambition is not merely to answer questions but to propose computational methods, test them against defined evidence, learn from failed candidates, repair them where possible and retain only demonstrated capabilities. It remains subject to human authority and independent qualification.

01 — NIKI's intended role

Given an authorized problem, NIKI may study allowed knowledge, generate candidate algorithms or Circuits and seek an executable answer. It is intended to work with FarmNodes, DataChips, BTDU, ENTITY and ADAM rather than replace them.

An articulate explanation or passing compilation is not sufficient. The important distinction is whether the new method performs against independent test cases and reproduces its result in a controlled environment.

02 — The research mandate

A responsible sponsor describes the actual problem, permitted data, desired output, independent acceptance criteria, computational budget, safety rules and qualified evaluator.

Without a clear oracle or measure of success, NIKI may produce interesting material that cannot be defended as a verified invention. The mandate must also prohibit unauthorized access, dangerous actions and leakage of protected information.

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03 — Candidate generation and preservation

The system should retain the original candidate, its input assumptions, resource conditions and test results. Any revision becomes a separately identifiable candidate so a later success does not erase an earlier failure.

A suggested repair that leaves the source unchanged or cannot execute due to missing dependencies is not a successful repair. A promising design remains a hypothesis until tested.

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04 — Independent tests and adversarial evaluation

A strong evaluation uses tests not simply memorized or supplied as worked examples to the inventing process. It should include boundary values, failure cases, alternative runtimes and resource restrictions appropriate to the task.

A perfect result over a bounded local oracle does not prove all possible inputs or transfer to a new problem class. Reviewers need to see test size, candidate identity, failed runs and whether test data were held out.

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05 — Learning from errors

A useful research cycle identifies where a proposal deviated from the expected result, forms an explanation and produces a meaningfully altered candidate or testable revision. Retaining unsuccessful approaches helps avoid repeating the same nonrepair.

The goal is evidence-based improvement, not unrestricted self-modification. Proposed changes must pass controlled admission gates before they affect production applications.

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06 — DataChips and Circuit Genomes

A validated NIKI method can be packaged as a versioned reusable capability with a clear input/output contract and associated evidence. A Circuit Genome can preserve the relationships among several such capabilities.

Promotion depends on compatibility, ownership, licence rights, and operator approval. A local success must not automatically provide access to unrelated devices, financial systems or personal data.

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07 — FarmNode collaboration

A research task may be worked on by one logical farm and challenged by another. Independent host or institutional review can test whether a candidate works outside its originating environment.

Multiple processes on the same desktop do not establish independent scientific replication. A valid external challenge must document what the reviewer actually reran or independently re-created.

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08 — The role of language models

A language model such as a Qwen-family system may help with reasoning, coding and idea generation, if its contribution is measured against the existing NIKI algorithm-invention baseline. It is a tool within the architecture, not a replacement for NIKI’s experiment, validation and ownership framework.

Model size does not demonstrate practical improvement. Hardware cost, reproducibility, measured algorithm quality and safety must govern whether a model is introduced.

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09 — Failures and bounded evidence

BTG’s public historical records include a sensor repair later passing all 65,536 cases of one local oracle and a distinct Granby candidate that produced 256 incorrect cases out of 65,536 in its recorded test. The two results address different tasks and cannot be conflated.

The existence of generated repair source is not the same as a qualified changed algorithm. Independent runtime and dependency qualification must accompany future claims of success.

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10 — Safety, intellectual property and release

New invention records should document provenance, permitted source materials, potential third-party rights, dangerous or prohibited uses, disclosure restrictions and lawful commercial classification.

NIKI should not grant itself rights, issue investment securities or decide that a technically functioning output is legally marketable. Responsible owners and independent reviewers must approve those actions.

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11 — Current status and the next evidence gate

The intended end-state includes reproducible invention, robust self-repair, independently qualified execution environments and release-ready reusable components. Public material shows meaningful bounded progress but not unrestricted general invention or fully qualified physical multi-node production.

This document is a public account of the programme, not a disclosure of confidential models, private algorithms, source repositories or production credentials.

Related manuals and use

The companion User & Operator Manual describes the decisions and reviews applicable to this system. References to proposed user flows are explicitly conditional on qualification of the relevant product release.

Updated 23 October 2026 · Editorial documentation, not proof of regulatory authorization or release readiness · Proprietary implementation withheld

Expanded system explanation

Question	Clear answer
Audience and purpose	NIKI — Cognitive Research & Invention is described for the public, institutional evaluators and prospective partners; this is not a software implementation reference.
What exists	An experimental research system intended to propose executable solutions, test them against independent rules, diagnose failure, and retain only qualified knowledge.

Question	Clear answer
Analogy	A research assistant in a supervised laboratory that must show working experiments and failed attempts, not just convincing answers.
Authority	The described process requires actual contracts, human authority and technical qualification at every relevant boundary.

Practical examples — what a reader should infer

Bounded success

An original sensor repair matched 65,536 of 65,536 local test cases in a specified WebAssembly experiment. That supports a narrow algorithm result, not a claim that NIKI can invent arbitrary correct software.

Bounded failure

The Granby research task previously reported 256 incorrect cases out of 65,536. Such results must remain visible and rejected or under repair, not silently counted as successes.

Reusable component

A candidate method may become a DataChip only after frozen-source tests, rights review, runtime qualification and authorized admission.

System lifecycle — each separate decision

1. Receive an authorized problem and success criteria

2. Search permitted prior knowledge and propose a candidate
3. Freeze the exact candidate before evaluation
4. Execute in a bounded test environment
5. Compare independent oracle and negative cases
6. Preserve failures, repairs and qualification decision

People, organizations and responsible roles

- Research sponsor defines objectives and lawful sources
- NIKI proposes candidates; proposals are not certifications
- Sandbox enforces execution restrictions
- Independent oracle/reviewer judges actual behaviour
- Rights/owner authority authorizes any governed retention or release

Where the system can fail

- Language model size alone does not establish algorithm correctness or autonomous discovery.
- A successful narrow experiment does not establish general invention reliability.
- Untrusted generated code needs sandbox containment and adversarial tests.
- Repair attempts producing unchanged code are not evidence of successful correction.

Evidence a serious evaluator would require

These are acceptance requirements, not claims that each artifact has been issued or independently certified:

1. Frozen candidate hashes and prompt/mandate references
2. Independent test oracle and holdout definitions
3. Raw pass/fail counts and observed counterexamples
4. Sandbox cgroup/rlimit/network negative evidence
5. Three-host replication if distributed claims made
6. Owner-signed governed admission and provenance

Terms explained in ordinary language

Term	Meaning
Oracle	A separate criterion or evaluator for deciding correctness.
Holdout	A test withheld from generating or fitting the solution.
Candidate	A proposed method not yet accepted as correct.
Promotion	Controlled admission of a validated capability into a reusable system.

How to read the current status accurately

The architecture and local engineering records are distinct from production operation. A named component or test figure cannot establish authorized commercial rights, fully qualified infrastructure or independently accepted financial activity. Read the dated corporate evidence register for each claimed version; ask for a scoped, access-controlled evidence bundle for decisions that rely on private implementation.

More detailed reading: User & Operator Manual · Non-source engineering evaluation · Regulatory assessment pathway · Complete cross-system example.